MICHAEL F. PÉREZ

Machine Learning & Computer Vision Engineer
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Machine learning and computer vision engineer with a Ph.D. in Computer Science from the University of Florida. I build and evaluate ML systems end to end. At Raft, I integrated computer vision data exploration and analysis into production software and built ML data and evaluation workflows. In my doctoral research, I developed video understanding models for action recognition, detection, segmentation, and retrieval, applied and fine-tuned vision-language models, and built the pipelines and tools around them, with an emphasis on efficiency and real-time inference.

EMPLOYMENT

Senior Engineer August 2026 – November 2026
Raft Tampa, FL

- Developed and integrated computer vision data exploration and analysis capabilities into a larger production software system, including dataset visualization, clustering, filtering, natural-language querying, and geospatial analytics.
- Built ML data management and evaluation workflows to support dataset curation, annotation review, model analysis, and debugging.
- Contributed to the development, testing, and integration of AI/ML capabilities, including object-detection experimentation and improvements to existing software systems.

Machine Learning Engineer Intern May – August 2025
CoVar Durham, NC

- Improved resolution of infrared video via open-source super-resolution model to classify distant objects.
- Developed and trained lightweight RGB object-detection models for real-time drone detection, targeting resource-constrained, on-device use.
- Built offline computer-vision evaluation pipelines spanning preprocessing, model inference, metric computation, and efficiency analysis.

Research Assistant August 2020 – August 2026
University of Florida Gainesville, FL

- Member of Ruiz Human-Computer Interaction Lab, Data Science Research Lab, and Graphics, Imaging and Light Measurement Lab.
- Conducted interdisciplinary research applying computer vision to AR, web, and neuroscience projects.
- Principal Investigators: Dr. Jaime Ruiz, Dr. Daisy (Zhe) Wang, Dr. Corey Toler-Franklin.

Teaching Assistant August 2020 – August 2026 (various semesters)
University of Florida Gainesville, FL

- TA for Analysis of Algorithms, Data Structures and Algorithms, Computational Structures for Computer Graphics, Operating Systems, Programming Fundamentals II, Performant Programming in Python, Deep Learning for Computer Graphics, and Algorithm Abstraction and Design.
- Graded assignments, developed automatic grading scripts, hosted discussions, and held office hours for hundreds of students.

Research Assistant, Research Intern June 2019 – May 2020
Florida Polytechnic University Lakeland, FL

- Applied computational methods to rhinoplasty research.
- Principal Investigator: Dr. Oguzhan Topsakal.

Student Educational Assistant June 2019 – May 2020
Database 1 and Software Engineering, Florida Polytechnic University Lakeland, FL

- Graded 50 assignments and 50 quizzes biweekly for Database 1 and Software Engineering.

TECHNICAL SKILLS

Machine Learning & Computer Vision: PyTorch, TensorFlow, OpenCV, scikit-learn, Hugging Face, NumPy, SciPy, pandas; vision-language models (LaViLa, LLaVA, Qwen)
MLOps & Data Tools: Kubernetes, Airflow, MLflow, Label Studio, FiftyOne
Programming Languages: Python, C++, SQL, JavaScript, C, Java, C#, F#, Julia, GLSL, HTML, CSS
Systems & Backend: Docker, FastAPI, Flask, Celery, Redis, Linux, Bash, SLURM/HPC, ROS, Git, CMake
Graphics & Applications: OpenGL, WebGL, MATLAB, Blender, LaTeX, Visual Studio
Operating Systems: Ubuntu, macOS, Windows
Spoken Languages: Bilingual (English and Spanish)

EDUCATION

Ph.D. in Computer Science August 2026
University of Florida, GPA: 3.77 Gainesville, FL

M.S. in Computer Science May 2023
University of Florida, GPA: 3.75 Gainesville, FL

B.S. in Computer Science May 2020
Florida Polytechnic University, GPA: 3.8 Lakeland, FL

CERTIFICATIONS

CompTIA Security+ October 2026

RESEARCH EXPERIENCE

Ph.D. Thesis: Efficient Video Action Detection 2025 – 2026
CISE Department, University of Florida Gainesville, FL

- Designed a causal, online (streaming) temporal action-detection model on state-space (Mamba) backbones for higher temporal accuracy and lower latency in long videos.
- On THUMOS, the model reaches 73.3 streaming mAP with a ResNet-50 backbone and 76.2 with DINOv2, matching the strongest verified strictly-causal detector at roughly 60% of its parameters and 2.5–2.9× its throughput, and runs in real time at 37.7 FPS.
- Built a unified framework to benchmark online, sliding-window, and multimodal video models on THUMOS and ActivityNet across mAP, latency, throughput, memory, and FLOPs, with distributed multi-GPU training on a SLURM-managed HPC cluster.
- First-author manuscript in preparation.

ECOLE Program – CReLeRI Team (DARPA-funded) July 2023 – May 2025
Ruiz HCI Lab and DSR Lab, CISE Department, University of Florida Gainesville, FL

- Developed and integrated UI and image/video analysis APIs and trained/tested action segmentation models for an explainable long-video analysis system; first author of its ACM Multimedia 2025 demo.
- Built the system's long-video analysis web tool: an HTML/JS/CSS front end and a FastAPI backend with Celery and Redis background tasks, containerized with Docker, with GPU compute offloaded to UF's HiPerGator cluster.
- Applied vision-language and LLM-based models, using LLaMA to parse Qwen-generated captions in the final system.

PTG Program – ENKIx Team (DARPA-funded) July 2023 – December 2024
Ruiz HCI Lab, CISE Department, University of Florida Gainesville, FL

- Trained and tested action recognition models, including fine-tuning the video-language model LaViLa, for a real-time AR task guidance system; achieved 34% F1-score in task recognition (mid-range results among DARPA performers) and conducted usability and cognitive load evaluations.

Animal Behavior Recognition (NSF-funded) August 2020 – July 2023
Graphics, Imaging, and Light Measurement Lab, CISE Department, UF Gainesville, FL

- Investigated adversarial approaches for modeling 3D motion and behavior in videos for applications in biomedical research.
- Conducted a survey of animal action recognition algorithms, trained and tested existing SOTA methods, and began implementing novel semi-supervised adversarial algorithms.

Rhinoplasty Research Group May 2019 – May 2020
Computer Science Department, Florida Polytechnic University Lakeland, FL

- Conducted a survey of over fifty research papers, documenting the facial landmarks and measurements relevant to the field of rhinoplasty.
- Collaborated with other students to develop a web interface for importing, exporting, and displaying feature point coordinates and measurement values onto 3D models of the human face.
- Achieved 95% agreement with Blender software and outperformed 2D methods in inter-/intra-rater reliability and usability metrics.

SELECTED PROJECTS

Gossip and Push-Sum Simulator September – October 2021
F#, Akka.NET github.com/michaelperez023/gossip-simulator

- Built a distributed-systems simulator modeling gossip and push-sum convergence across full, line, 3D, and imperfect-3D topologies, scaling past 2 million nodes.

Distributed Twitter Clone November – December 2021
F#, Akka.NET, Suave; with one other student github.com/michaelperez023/twitter-clone-part2

- Built a Twitter clone on the actor model, supporting registration, tweets, retweets, hashtags, and mentions (Distributed Operating Systems course project).

Portfolio Website 2026
React, Vite michaelfperez.com

- Built an interactive, scrubbable timeline of my work with a question-answering feature that runs entirely in the browser (BM25 keyword retrieval plus sentence embeddings), with no backend.

HONORS AND AWARDS

GenerationNext Grant January 2021 – August 2026

- Multifaceted cohort-based support system to aid Ph.D. students in the UF CISE department.

Graduate School Preeminence Award September 2020 – August 2026

- Highly competitive research assistantship stipend offered to UF CISE's strongest Ph.D. applicants.

L3Harris Corporation Graduate Fellowship April 2022

- Scholarship awarded to five UF CISE graduate students based on academic performance (e.g., GPA), research activity (e.g., publications and awards), and professional services.

PROFESSIONAL ACTIVITIES & SERVICE

Peer Reviewer, ICXR 2026 — August 2026

Peer Reviewer, ACM International Conference on Multimedia Retrieval (ICMR) — March 2026

Peer Reviewer, ACM Multimedia — May – June 2025

Co-Chair, UF Human-Centered Data Science Seminar — Spring 2025

Peer Reviewer, IEEE Conference on Virtual Reality and 3D User Interfaces — October – November 2024

Panelist — February 2023
Black Male Computer Scientist Panel, Reichert House Youth Academy — Gainesville, FL

- Engaged with high schoolers to provide college and career advice.

PUBLICATIONS

Journal Articles

1. R. Shahriari, Y. Yang, D. Tamboli, M. F. Pérez, Y. Zha, J. Hou, M. Deng, E. D. Ragan, J. Ruiz, D. Z. Wang, Z. Hu, and E. Xing, "**MuCHEx: A Multimodal Conversational Debugging Tool for Interactive Visual Exploration of Hierarchical Object Classification**," *IEEE Computer Graphics and Applications*, vol. 45, no. 6, pp. 76–88, 2025. doi: 10.1109/MCG.2025.3598204.
2. M. M. Celikoyar, M. F. Pérez, M. I. Akbaş, and O. Topsakal, "**Facial Surface Anthropometric Features and Measurements with an Emphasis on Rhinoplasty**," *Aesthetic Surgery Journal*, vol. 42, no. 2, pp. 133–148, 2022. doi: 10.1093/asj/sjab190.
3. O. Topsakal, M. İ. Akbaş, B. S. Smith, M. F. Pérez, E. C. Guden, and M. M. Celikoyar, "**Evaluating the agreement and reliability of a web-based facial analysis tool for rhinoplasty**," *International Journal of Computer Assisted Radiology and Surgery*, vol. 16, no. 8, pp. 1381–1391, 2021. doi: 10.1007/s11548-021-02423-z.
4. O. Topsakal, M. İ. Akbaş, D. Demirel, R. Nunez, B. S. Smith, M. F. Pérez, and M. M. Celikoyar, "**Digitizing rhinoplasty: a web application with three-dimensional preoperative evaluation to assist rhinoplasty surgeons with surgical planning**," *International Journal of Computer Assisted Radiology and Surgery*, vol. 15, no. 11, pp. 1941–1950, 2020. doi: 10.1007/s11548-020-02251-7.

Conference Proceedings

1. M. F. Pérez, Y. Yang, Y. Zha, E. Ma, D. Tamboli, H. Ma, R. Shahriari, V. Pathak, D. Kasinets, R. Venkatakrishnan, D. Z. Wang, J. Ruiz, E. D. Ragan, Z. Hu, E. Xing, and J.-Y. Zhu, "**CReLeRI: Explainable, Concept-centric, Representation, Learning, Reasoning, and Interaction Video Analysis System**," in *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 13528–13530, 2025 (demo). doi: 10.1145/3746027.3754479.
2. M. F. Pérez, D. Kalika, and P. Torrione, "**Super-Resolution for Improving Vehicle Recognition at Large Stand-offs**," in *Military Sensing Symposium (MSS) (BSD, Materials & Detectors, and Passive Sensors) Conference*, March 2–6, 2026.
3. O. Topsakal, M. İ. Akbaş, B. S. Smith, M. F. Pérez, E. C. Guden, and M. M. Celikoyar, "**Evaluating Intra and Inter Reliability of a Web-Based Facial Analysis Tool for Rhinoplasty**," in *Computer Assisted Radiology and Surgery (CARS) Conference*, Munich, Germany, June 21–25, 2021.
4. O. Topsakal, M. İ. Akbaş, D. Demirel, R. Nunez, B. S. Smith, M. F. Pérez, and M. M. Celikoyar, "**Digitizing Rhinoplasty: A web application for three-dimensional preoperative planning**," in *Computer Assisted Radiology and Surgery (CARS) Congress*, Munich, Germany, June 23–27, 2020.

Manuscripts Under Review

1. R. Calvo, C. Bowers, M. F. Pérez, A. Barquero, and J. Ruiz, "**The Effects of Multiple Companion Agents on User Experience and Mental Workload**," under review at the *ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)*.

Manuscripts in Preparation

1. M. F. Pérez, A. Barquero, R. Calvo, D. Moore, M. Zolotas, N. Grimaldi, Y. Liu, E. Patterson, D. Bista, N. Regmi, R. Venkatakrishnan, L. Anthony, K. Brown, D. Wozniak, I. Wang, A. Tompkins, J. Fairbanks, T. Padir, M. Zahabi, D. Kaber, and J. Ruiz, "**ENKIx: An Intelligent AR Task Guidance System with Cognitive Load-Adaptive Capabilities**." Ph.D. thesis work; first-author manuscript in preparation.
2. M. F. Pérez, D. Z. Wang, and J. Ruiz, "**Temporal Granularity as a Design Variable in Zero-Shot Video Retrieval**." Manuscript in preparation.

Preprints

1. M. F. Pérez and C. Toler-Franklin, "**CNN-Based Action Recognition and Pose Estimation for Classifying Animal Behavior from Videos: A Survey**," arXiv preprint arXiv:2301.06187, 2023.

REFERENCES

Dr. Jaime Ruiz, Associate Professor (Ph.D. co-supervisor) jaime.ruiz@ufl.edu
Computer and Information Science and Engineering Department, University of Florida Gainesville, FL

Dr. Daisy (Zhe) Wang, Professor (Ph.D. co-supervisor) daisyw@cise.ufl.edu
Computer and Information Science and Engineering Department, University of Florida Gainesville, FL

Dr. Corey Toler-Franklin, Assistant Professor (former Ph.D. supervisor) ctolerfr@barnard.edu
Computer Science Department, Barnard College New York, NY

Dr. Oguzhan Topsakal, Associate Professor (undergraduate research supervisor) topsakal@usf.edu
Computer Science Department, University of South Florida Tampa, FL